

10. THE PROFILE IN ARBITRARY CHARACTERISTIC, AND THE CHARACTERISTIC-ZERO PICTURE

Addendum (27 September 2026). Sections 1–9 treat $G = \mathbb{Z}/p^n$ in characteristic p . This section records what the definition of the profile gives for an arbitrary finite group G over an arbitrary field k , and what it looks like in characteristic 0 (more generally, in characteristic prime to $|G|$), where the mechanisms are entirely different: instead of Artin–Schreier layers, the profile is governed by Sylow divisibility (Theorem 10.10) and by the roots-of-unity defect $\mathrm{ed}_k(G) - \mathrm{ed}_{k(\mu_e)}(G)$ (Corollary 10.5). Statements carry the tags of Section 1; in addition, §10.6 says explicitly which statements are rewrites of results in the literature and which are new. As in the rest of the note, literature is quoted without re-verification and the corresponding statements are tagged [cited].

10.1. Conventions. k is any field, G a finite group (constant group scheme over k), $F \supseteq k$ a field, $\tau: T \rightarrow \mathrm{Spec} F$ a G -torsor. The classifying homomorphism $\varphi_\tau: \Gamma_F \rightarrow G$ is defined up to conjugation, its image $H = \mathrm{im} \varphi_\tau$ up to conjugacy, and $T \cong \coprod_{G/H} \mathrm{Spec} E$ with E/F Galois with group H (the *splitting field* of τ). We use, in every characteristic: (B1) $\mathrm{ed}_k(\tau \otimes_F F') \leq \mathrm{ed}_k(\tau)$; (B2) $\mathrm{ed}_k(\tau) \geq \mathrm{ed}_{k'}(\tau)$ for $k \subseteq k' \subseteq F$; (B3) $\mathrm{ed}_k(\tau) = 0$ iff τ descends to the algebraic closure of k in F (for k algebraically closed: iff τ is split; for general k this is *not* “descends to k ”: $k = \mathbb{Q}$, $F = \mathbb{Q}(i)(t)$, $\tau = F(\sqrt{1+i})$ has $\mathrm{ed}_{\mathbb{Q}} = 0$ but is not $F(\sqrt{a})$ with $a \in \mathbb{Q}$, since $N_{\mathbb{Q}(i)/\mathbb{Q}}(1+i) = 2$ is not a square); (B4) Lemma 4.1(i), purely transcendental base change does not change ed_k . For a prime q , $\mathrm{ed}_k(\tau; q) = \min\{\mathrm{ed}_k(\tau \otimes F') : F'/F \text{ finite separable, } q \nmid [F' : F]\}$ and $\mathrm{ed}_k(G; q) = \sup_\tau \mathrm{ed}_k(\tau; q)$ [29, (2.5)]. The generic torsor $\tau_{\mathrm{gen}}: \mathrm{Spec} L \rightarrow \mathrm{Spec} K$, $L = k(V)$, $K = L^G$, is taken for a faithful representation V (or any faithful generically free G -variety with $k(V)/k$ regular); $\mathrm{ed}_k(\tau_{\mathrm{gen}}) = \mathrm{ed}_k(G)$ [18, 27]. The profile $\mathrm{ed}_k^{[d]}(\tau)$ and the jump degrees $d_j(\tau)$ are as in Definition 3.1.

10.2. Formal properties, valid in every characteristic.

Proposition 10.1 (shape). *Let τ be a G -torsor over $F \supseteq k$.*

- (a) $d \mapsto \mathrm{ed}_k^{[d]}(\tau)$ is non-increasing with values in $\{0, \dots, \mathrm{ed}_k(\tau)\}$, $\mathrm{ed}_k^{[1]}(\tau) = \mathrm{ed}_k(\tau)$, and $d_{\mathrm{ed}(\tau)} = 1 \leq d_{\mathrm{ed}(\tau)-1} \leq \dots \leq d_0 \leq |\mathrm{im} \varphi_\tau|$.
- (b) If $[F' : F] = e$ and $d \geq 1$ then $\mathrm{ed}_k^{[de]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau \otimes F') \leq \mathrm{ed}_k^{[d]}(\tau)$; hence $d_j(\tau) \leq [F' : F] d_j(\tau \otimes F')$ for every finite F'/F and every j .
- (c) $\mathrm{ed}_k^{[d]}(\tau) = \min\{j : d_j(\tau) \leq d\}$: the jump degrees determine the profile.
- (d) If $k \subseteq k' \subseteq F$ then $\mathrm{ed}_{k'}^{[d]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau)$ for all d .
- (e) $\mathrm{ed}_k^{[d]}(\tau) = 0$ iff some F'/F of degree $\leq d$ makes $\tau \otimes F'$ descend to the algebraic closure of k in F' ; for k algebraically closed, iff τ is split by an extension of degree $\leq d$.

[proved]

Proof. (a) Over the splitting field E , $[E : F] = |H|$, the torsor is split, so $d_0 \leq |H|$; the rest is (B1). (b) is Proposition 3.2(b) verbatim; if $[F'' : F] = d_j(\tau \otimes F')$ realises $\mathrm{ed}(\tau \otimes F'') \leq j$ then $[F'' : F] = [F' : F] d_j(\tau \otimes F')$. (c) is a restatement, (d) is (B2) applied to each $\tau \otimes F'$, (e) is (B3). $\square$

Proposition 10.2 (the last jump degree). *Let $H = \mathrm{im} \varphi_\tau$ and E the splitting field of τ .*

- (a) $d_0(\tau) \leq \min\{[F' : F] : \tau \otimes F' \text{ split}\} = |H|$, with equality if k is algebraically closed.
- (b) Let k be perfect and E/k regular. If $[F' : F] < \infty$ and $\tau \otimes F'$ descends to the algebraic closure k_1 of k in F' , then $[F' : F] \geq |H| \cdot [k_1 : k]$. In particular $d_0(\tau) = |H|$.

- (c) For τ_{gen} over a perfect k : $d_0(\tau_{\text{gen}}) = |G|$, and $\text{ed}_k(\tau_{\text{gen}} \otimes F') = 0$ forces $[F' : K] \geq |G| [k_1 : k]$.

[proved]

Proof. (a) $T \otimes_F F'$ is split iff E embeds F -linearly into F' , which forces $[F' : F] \geq |H|$; $F' = E$ works; for k algebraically closed use (B3). (b) Let $\tau \otimes F' \cong \tau_0 \otimes_{k_1} F'$ with τ_0 over k_1 , of splitting field l/k_1 Galois with group H_0 . Comparing total spaces inside an algebraic closure of F' we may arrange $EF' = lF'$. Since l/k is finite separable and E/k regular, $l \otimes_k E$ is a field and $[lE : E] = [l : k] = |H_0| [k_1 : k]$; as $lE \subseteq EF'$,

$$|H_0| [k_1 : k] |H| = [lE : E][E : F] \leq [EF' : F] = [lF' : F'] [F' : F] \leq |H_0| [F' : F].$$

- (c) $E = L$ is regular over k and $H = G$. □

Remark 10.3. Without regularity $d_0 < |H|$ is possible ($k = \mathbb{Q}$, $\tau = \mathbb{Q}(t)(\sqrt{2})$ is constant, $d_0 = 1$). Proposition 10.2(c) is the general form of Proposition 3.2(c). For k algebraically closed, $\text{ed}^{[d]}(\tau) \geq 1$ for $d < |H|$: the last slot of the profile is the only one understood in general.

Proposition 10.4 (base-field extension bound). *Let k'/k be finite of degree d . Then $\text{ed}_k^{[d]}(\tau) \leq \text{ed}_{k'}(G)$. More precisely, every field factor F' of $F \otimes_k k'$ has $[F' : F] \leq d$ and $\text{ed}_k(\tau \otimes F') \leq \text{ed}_{k'}(\tau \otimes F') \leq \text{ed}_{k'}(G)$.* [proved]

Proof. $F' \supseteq k'$ and $[F' : F] \leq d$. A descent $\tau \otimes F' \cong \tau_0 \otimes_{F_0} F'$ with $k' \subseteq F_0 \subseteq F'$ and $\text{trdeg}_{k'} F_0 \leq \text{ed}_{k'}(G)$ is also a descent over k , and $\text{trdeg}_k F_0 = \text{trdeg}_{k'} F_0$ since k'/k is algebraic. □

Corollary 10.5 (roots-of-unity defect). *Let e be the exponent of G , $\text{char } k \nmid e$, and $d_e = [k(\mu_e) : k]$. For every G -torsor τ over every $F \supseteq k$,*

$$\text{ed}_k^{[[F(\mu_e):F]]}(\tau) \leq \text{ed}_k^{[d_e]}(\tau) \leq \text{ed}_{k(\mu_e)}(G).$$

In particular: (i) if G is abelian, $\text{ed}_k^{[d_e]}(\tau) \leq \text{rank } G$ [19]; (ii) if G is a p -group and $\text{char } k \neq p$, $\text{ed}_k^{[p-1]}(\tau) \leq \text{ed}_k^{[[k(\mu_p):k]]}(\tau) \leq \text{ed}_{k(\mu_p)}(G)$, the minimal dimension of a faithful representation of G over $k(\mu_p)$ [24]. [proved; the values are cited]

Remark 10.6. For τ_{gen} and $F' = K(\mu_e)$ the torsor $\tau_{\text{gen}} \otimes F'$ is the generic torsor over $k(\mu_e)$, so $\text{ed}_{k(\mu_e)}(\tau_{\text{gen}} \otimes K(\mu_e)) = \text{ed}_{k(\mu_e)}(G)$ exactly, but whether ed_k of this torsor equals $\text{ed}_{k(\mu_e)}(G)$ is not known to us in general. [open]

10.3. Sylow divisibility of the jump degrees.

Lemma 10.7 (q -closure). *Let q be a prime, $S \leq \Gamma_F$ a Sylow pro- q subgroup and $F^{(q)} = (F^{\text{sep}})^S$. Then $F^{(q)}$ is a union of finite subextensions of degree prime to q , every finite separable extension of $F^{(q)}$ has q -power degree, and $\text{ed}_k(\tau; q) = \text{ed}_k(\tau \otimes F^{(q)})$ [29, (2.5)]. If F'/F is finite separable with $q \nmid [F' : F]$, then there is an F -embedding $F' \hookrightarrow F^{(q)}$, and hence $\text{ed}_k(\tau \otimes F') \geq \text{ed}_k(\tau; q)$.* [cited; last statement proved]

Proof. $F' \otimes_F F^{(q)}$ is a product of finite separable extensions of $F^{(q)}$, each of q -power degree, whose degrees sum to $[F' : F]$; if none had degree 1 the sum would be divisible by q . So some factor is $F^{(q)}$, giving the embedding, and (B1) gives the inequality. (For a *given* embedding $F' \subseteq F^{\text{sep}}$ the compositum $F'F^{(q)}$ need not be $F^{(q)}$: $F = \mathbb{Q}$, $q = 2$, $F' = \mathbb{Q}(\sqrt[3]{2})$ has $F' \otimes_{\mathbb{Q}} \mathbb{Q}^{(2)} \cong \mathbb{Q}^{(2)} \times \mathbb{Q}^{(2)}(\sqrt{-3})$.) □

Lemma 10.8 (purely inseparable invariance). *Let $\text{char } k = p$, k perfect, G any finite group, F'/F finite purely inseparable. Then $\text{ed}_k(\tau \otimes F') = \text{ed}_k(\tau)$; consequently $\text{ed}_k^{[d]}(\tau)$ is the minimum over separable F'/F of degree $\leq d$.* [proved]

Proof. As in Proposition 3.6, with $\sigma: F^{1/p^m} \rightarrow F$, $x \mapsto x^{p^m}$; the p^m -power map extends to $F^{1/p^m} F^{\text{sep}} \rightarrow F^{\text{sep}}$ and commutes with all automorphisms, so the induced isomorphism $\Gamma_{F^{1/p^m}} \rightarrow \Gamma_F$ is the canonical one and the classifying homomorphism of $\sigma_*(\tau \otimes F^{1/p^m})$ is φ_τ . This recovers Proposition 3.6 without Witt vectors. $\square$

Proposition 10.9. *For every prime q , $\text{ed}_k(\tau_{\text{gen}}; q) = \text{ed}_k(G; q)$. [cited: [29, 27]; sketch below]*

Sketch. “ $\leq$ ” is the definition. For “ $\geq$ ” let $e = \text{ed}_k(\tau_{\text{gen}}; q)$, realised by a prime-to- q extension K'/K and a descent to K_0 , $\text{trdeg}_k K_0 \leq e$; spread out to a G -torsor $Z \rightarrow B$, $\dim B \leq e$, and to a G -equivariant rational map $W' \dashrightarrow Z$ from the normalisation W' of V in $L \otimes_K K'$, finite of degree $[K' : K]$ over V and étale over a dense G -stable open $V_1 \subseteq V$. Given any torsor τ over an infinite F , the twist ${}^\tau V$ is an affine space over F , so a free F -point in ${}^\tau V_1$ exists and gives an equivariant $f: T \rightarrow V_1$; the pullback $T \times_V W' \rightarrow T$ is finite étale of degree $[K' : K]$, hence $\cong T \times_F A$ for a finite étale F -algebra A of that degree. Some field factor F' of A has $q \nmid [F' : F]$ and $T_{F'} \rightarrow W' \dashrightarrow Z$ is equivariant into a free G -variety, so $\text{ed}_k(\tau \otimes F') \leq \dim B \leq e$ by Lemma 2.3 (valid in any characteristic, Theorem 10.13). $\square$

Theorem 10.10 (Synd divisibility). *Let $\text{char } k = 0$, or k perfect (then use Lemma 10.8). Let τ be a G -torsor over F .*

- (a) *For every prime q and every $j < \text{ed}_k(\tau; q)$, q divides $d_j(\tau)$: every F'/F with $\text{ed}_k(\tau \otimes F') \leq j$ has $q \mid [F' : F]$.*
- (b) *Every prime q with $\text{ed}_k(G; q) > j$ divides $d_j(\tau_{\text{gen}})$; hence $d_j(\tau_{\text{gen}}) \geq \prod_{q: \text{ed}_k(G; q) > j} q$.*
- (c) *For every $d \geq 1$, $\text{ed}_k^{[d]}(\tau_{\text{gen}}) \geq \min_{1 \leq d' \leq d} \max_{q \nmid d'} \text{ed}_k(G; q) \geq \max_{q > d} \text{ed}_k(G; q)$ (q prime).*
- (d) *The prime-to- q profile $\text{ed}_k^{[d]'}(\tau) = \min\{\text{ed}_k(\tau \otimes F') : [F' : F] \leq d, q \nmid [F' : F]\}$ is non-increasing with terminal value $\text{ed}_k(\tau; q)$, reached at a finite prime-to- q degree; for τ_{gen} the terminal value is $\text{ed}_k(G; q)$.*

[proved, granting Proposition 10.9]

Proof. (a) is Lemma 10.7; (b) adds Proposition 10.9; (c) a witness for $\text{ed}^{[d]}$ has some degree $d' \leq d$ and (a) applies to every $q \nmid d'$; (d) is Lemma 10.7 together with the finiteness argument of Proposition 3.7(a). $\square$

Remark 10.11. (i) In the language of Theorem 4.2 and Remark 4.5, (a) is Reichstein–Scavia’s Theorem 10.1 applied to the correspondence of degree prime to q furnished by a free closed point; the proof above via the q -closure is the same mechanism. (ii) Consistency with Sections 3–9: for $G = \mathbb{Z}/p^n$ in characteristic p , $\text{ed}_k(G; p) \leq 1$ and $\text{ed}_k(G; q) = 0$ for $q \neq p$, so Theorem 10.10 only says $p \mid d_0$, consistent with $d_0 = p^n$; it says nothing about $d_1, \dots, d_{n-1}$, which is the point of this note. In characteristic 0 the constraint is much stronger.

Proposition 10.12 (q -groups). *Let G be a q -group of order q^n , $\text{char } k \neq q$ (k perfect if positive), $c = [k(\mu_q) : k]$. Then: (a) $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G; q)$ for $c \leq d \leq q - 1$, and $\text{ed}_k(G; q) \leq \text{ed}_k^{[d]}(\tau_{\text{gen}}) \leq \text{ed}_k(G)$ for $d < c$; (b) $\text{ed}_k(G; q) = \text{ed}_{k(\mu_q)}(G)$ is the minimal dimension of a faithful $k(\mu_q)$ -representation [24]; (c) $q \mid d_j(\tau_{\text{gen}})$ for every $j < \text{ed}_k(G; q)$, and $d_0 = q^n$; (d) if $\mu_q \subset k$: $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G)$ for $d < q$ and all steps of the profile occur at multiples of q , i.e. $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k^{[q \lfloor d/q \rfloor]}(\tau_{\text{gen}})$ for $d \geq q$. [proved, modulo [24] and $\text{ed}(\cdot; q)$ being invariant under prime-to- q extensions of k]*

Proof. Lower bounds: every degree $< q$ is prime to q (Theorem 10.10). Upper bound for $d \geq c$: Corollary 10.5 and $\text{ed}_{k(\mu_q)}(G) = \text{ed}_{k(\mu_q)}(G; q) = \text{ed}_k(G; q)$ [24, 27]. (c) is Theorem 10.10(b) and Proposition 10.2(c); (d) is (a)+(c) with $c = 1$. $\square$

Compare Proposition 3.7: in characteristic p the prime-to- p tail of a p -group collapses to ≤ 1 ; in characteristic $\neq q$ the prime-to- q tail of a q -group is *constant*, equal to $\text{ed}_k(G; q)$.

10.4. Geometric meaning and induced torsors in every characteristic.

Theorem 10.13. *Theorem 4.2 and Corollary 4.4 hold for every finite group G , every field k and every characteristic: $\text{ed}_k^{[d]}(\tau) \leq m$ iff some quasi-projective generically free primitive G -variety Z of dimension $\leq m$ has a closed point of degree $\leq d$ in the free locus of ${}^\tau Z$, iff there are such Z , F'/F of degree $\leq d$ and a G -equivariant morphism $T_{F'} \rightarrow Z^{\text{free}}$. For τ_{gen} this is a G -correspondence $V \rightsquigarrow Z$ of degree $\leq d$ into Z^{free} . [proved]*

Proof. The proof of Theorem 4.2 uses only: the existence of the twist ${}^\tau Z = (T \times_k Z)/G$ as a quasi-projective F -scheme compatible with base change and open immersions [21, §3]; the identification of F -points of ${}^\tau Z$ with equivariant maps $T \rightarrow Z_F$ (elementary for finite G); the fact that $Z^{\text{free}} \rightarrow Z^{\text{free}}/G$ is a G -torsor (SGA 1, V, finite group acting freely); and spreading out. None of these depends on the characteristic or on G being a cyclic p -group. $\square$

Lemma 10.14 (induced torsors). *For finite groups $H \leq G$ and an H -torsor τ_H over F , $\text{ed}_k(\text{Ind}_H^G \tau_H) = \text{ed}_k(\tau_H)$. [proved]*

Proof. “ $\leq$ ”: descend and induce. “ $\geq$ ”: reduce to T_H a field (if $T_H = \text{Ind}_H^H T'$ with T' a field, then $\text{Ind}_H^G \tau_H = \text{Ind}_H^G \tau'$ and $\text{ed}(\tau_H) \leq \text{ed}(\tau')$). Then the argument of Proposition 3.9 goes through for non-abelian G : a descent $T_0 \rightarrow \text{Spec } F_0$ has a connected component T'_0 with stabiliser H_0 ; the components of $T'_0 \otimes_{F_0} F$ are among the gT_H and are permuted transitively by H_0 , so after conjugating H we may assume $T_H \subseteq T'_0 \otimes_{F_0} F$ and $H \leq H_0$. With $F_1 = (T'_0)^H$ and $F_1 \otimes_{F_0} F = \prod F_i$, the factor $T'_0 \otimes_{F_1} F_{i_0}$ containing T_H is an H -torsor over F_{i_0} on whose components H acts transitively; being H -stable, T_H is that factor, and $F_{i_0} = T_H^H = F$. So τ_H descends to F_1 , $\text{trdeg}_k F_1 = \text{trdeg}_k F_0$. $\square$

Corollary 10.15 (subgroup profile). *For every subgroup $H' \leq G$: $\text{ed}_k^{[[G:H']]}(\tau_{\text{gen}}) \leq \text{ed}_k(\tau_{\text{gen}} \otimes L^{H'}) = \text{ed}_k(H')$. Hence $d_j(\tau_{\text{gen}}) \leq \min\{[G:H'] : \text{ed}_k(H') \leq j\}$. [proved]*

Proof. $\tau_{\text{gen}} \otimes L^{H'} = \text{Ind}_{H'}^G(L/L^{H'})$, and $L/L^{H'}$ is the generic H' -torsor of $V|_{H'}$ over k ; apply Lemma 10.14. $\square$

This is the general form of Proposition 3.9 and of the root-field reductions of Example 10.20.

10.5. Examples in characteristic zero.

Example 10.16 ($\mathbb{Z}/n$). Let $\mu_n \subset k$, $\text{char } k \nmid n$. The generic torsor $K(\sqrt[n]{a})/K$, $K = k(a)$, has $\text{ed} = 1$ and $d_0 = n$ (Proposition 10.2), so its profile is $(1, \dots, 1, 0)$. [proved]

Theorem 10.17 ($(\mathbb{Z}/2)^n$). *Let k be algebraically closed of characteristic $\neq 2$, $G = (\mathbb{Z}/2)^n$, $F = K = k(a_1, \dots, a_n)$, $\tau_{\text{gen}} = (a_1, \dots, a_n) \in H^1(F, \mathbb{Z}/2)^n$.*

- (a) $\text{ed}^{[2^j]}(\tau_{\text{gen}}) \leq n - j$ for $0 \leq j \leq n$, and $d_0 = 2^n$.
- (b) Every F'/F of odd degree has $\text{ed}(\tau_{\text{gen}} \otimes F') = n$; all d_j , $j < n$, are even.
- (c) For every quadratic F'/F , $\text{ed}(\tau_{\text{gen}} \otimes F') \geq n - 1$. Hence $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$, $d_{n-1} = 2$, and $d_{n-2} = 4$ for $n \geq 2$.

- (d) $\text{ed}^{[d]} = 1$ for $2^{n-1} \leq d < 2^n$. Complete profiles: $(1, 0)$ for $n = 1$; $(2, 1, 1, 0)$ for $n = 2$; $(3, 2, 2, 1, 1, 1, 0)$ for $n = 3$.

[proved; inputs cited]

Proof. (a) Over $F(\sqrt{a_1}, \dots, \sqrt{a_j})$ the class is $(1, \dots, 1, a_{j+1}, \dots, a_n)$. (b) $\text{ed}_k(G; 2) = \text{ed}_k(G) = n$ [19, 24]; apply Theorem 10.10. (d) from (a), (b), (c) and Proposition 10.2.

(c) Write $(a_I) = \bigcup_{i \in I} (a_i) \in H^{|I|}(F, \mathbb{Z}/2)$. Suppose $\text{ed}(\tau_{\text{gen}} \otimes F') \leq n-2$ for $F' = F(\sqrt{f})$. The class descends to F_0 with $\text{trdeg}_k F_0 \leq n-2$, so $H^{n-1}(F_0, \mathbb{Z}/2) = 0$ [30, II.4.2] and $(a_I)_{F'} = 0$ for all $|I| = n-1$. By the Arason exact sequence $H^{r-1}(F) \xrightarrow{(f) \cup} H^r(F) \xrightarrow{\text{res}} H^r(F(\sqrt{f}))$ [17], $(a_{[n] \setminus i}) \in (f) \cup H^{n-2}(F)$ for all i . We show by induction on m that for $F_m = k(a_1, \dots, a_m)$ there is no $g \in F_m^\times$ with $(a_{[m] \setminus i}) \in (g) \cup H^{m-2}(F_m)$ for all $i \in [m]$. For $m = 1$: $(a_\emptyset) = 1 \neq 0$ in H^0 . For $m \geq 2$ pass to $\hat{F} = F_{m-1}((a_m))$, where $H^r(\hat{F}) = H^r(F_{m-1}) \oplus H^{r-1}(F_{m-1}) \cup (a_m)$ and units have $(u) = (\bar{u})$ [30, II.4.3]; the classes (a_I) restrict injectively (their residues are nonzero by induction on the residue maps). Write $g = a_m^\varepsilon u$ and the given $x_i = y_i + z_i \cup (a_m)$. If $\varepsilon = 1$, take $i = m$: comparing the two summands of $(a_{[m-1]}) = (a_m u) \cup x_m$ gives $y_m = (\bar{u}) \cup z_m$ and $(a_{[m-1]}) = (\bar{u}) \cup (\bar{u}) \cup z_m = 0$ (as $-1 \in k^{\times 2}$), a contradiction. If $\varepsilon = 0$, for $i < m$ the second summands give $(a_{[m-1] \setminus i}) = (\bar{u}) \cup z_i$ for all $i \in [m-1]$, contradicting the induction hypothesis with $g = \bar{u}$. So no quadratic extension lowers ed by two; with (a), (b) this gives $\text{ed}^{[2]} = \text{ed}^{[3]} = n-1$, $d_{n-1} = 2$, $d_{n-2} = 4$. $\square$

Remark 10.18. Whether $\text{ed}^{[2^j]}(\tau_{\text{gen}}) = n-j$ for $2 \leq j \leq n-2$ is open; the first case is $n = 4$: is there a quartic extension of $k(a_1, \dots, a_4)$ splitting all six quaternion algebras (a_i, a_j) ? Non-Galois quartics have no kernel description as clean as Arason's, which is what makes $\text{ed}^{[4]}$ hard. [open]

Example 10.19 ($\mathbb{Z}/p$ over $\mathbb{Q}$). Let p be an odd prime and τ any $\mathbb{Z}/p$ -torsor over $F \supseteq \mathbb{Q}$. By Corollary 10.5 and Example 10.16, $\text{ed}_{\mathbb{Q}}^{[F(\mu_p):F]}(\tau) \leq 1$, so $d_1(\tau) \leq p-1$ for every such torsor, although $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p) \geq 2$ for $p \geq 5$ [26]: $\text{ed}_k(G) = 1$ in characteristic 0 iff $G \hookrightarrow \text{PGL}_2(k)$, and $\mathbb{Z}/p \not\hookrightarrow \text{PGL}_2(\mathbb{Q})$ for $p \geq 5$ since $\zeta_p + \zeta_p^{-1} \notin \mathbb{Q}$. The profile of the generic $\mathbb{Z}/p$ -torsor over $\mathbb{Q}$ thus drops to 1 at a degree dividing $p-1$, prime to p , consistent with Theorem 10.10 since $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p; p) = 1$. Whether $d_1(\tau_{\text{gen}}) = [\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = (p-1)/2$ is open. [proved from [26]]

Example 10.20 (symmetric groups). Let $\text{char } k = 0$, $L = k(x_1, \dots, x_n)$, $F = K = k(s_1, \dots, s_n)$, so τ_{gen} is the generic degree- n étale algebra. For $F_m = F(x_1, \dots, x_m)$, $[F_m : F] = n(n-1) \cdots (n-m+1)$, Corollary 10.15 and (B4) give

$$\text{ed}_k(\tau_{\text{gen}} \otimes F_m) = \text{ed}_k(S_{n-m}), \quad \text{so} \quad \text{ed}_k^{[n(n-1) \cdots (n-m+1)]}(\tau_{\text{gen}}) \leq \text{ed}_k(S_{n-m}).$$

With $\text{ed}_k(S_n; q) = \lfloor n/q \rfloor$ [28], Theorem 10.10 gives $\text{ed}^{[d]}(\tau_{\text{gen}}) \geq \lfloor n/q_d \rfloor$ for the least prime $q_d > d$, so the profile of the generic étale algebra decays no faster than $\sim n/(2d)$, whereas the root-field bound needs degree $\sim n^m$ for a drop of m . For S_6 (with $\text{ed}_k(S_6) = 3$, $\text{ed}_k(S_5) = \text{ed}_k(S_4) = 2$, $\text{ed}_k(S_3) = 1$ [19], and $\text{ed}_k(A_6) = 3$ [cited from memory]): $\text{ed}^{[1]} = 3$; $\text{ed}^{[d]} \geq 2$ for $2 \leq d \leq 5$; $\text{ed}^{[6]} \leq 2$, $\text{ed}^{[120]} \leq 1$, $d_0 = 720$; $2 \mid d_2$, $6 \mid d_1$; so $d_2 \in \{2, 4, 6\}$ and $d_1 \in 6\mathbb{Z} \cap [6, 120]$. The discriminant field L^{A_6} is *not* a witness for $d_2 = 2$, since over it $\text{ed} = \text{ed}_k(A_6) = 3$. [proved from cited inputs]

10.6. Provenance: rewrites versus new statements.

Statement	Status relative to the literature
Prop. 10.1 (shape)	Rewrite of Prop. 3.2 for arbitrary G, k ; formal.
Prop. 10.2 (last jump)	New but elementary; (a) is folklore, (b) the regularity refinement is new.
Prop. 10.4, Cor. 10.5	New (elementary); the values quoted are [19, 24].
Lemma 10.7	Rewrite of [29, (2.5)]; the embedding argument is standard.
Lemma 10.8	Rewrite of Prop. 3.6 in a form valid for every finite G (no Witt vectors).
Prop. 10.9	Known: [29, 27] (ed_q of a versal object).
Thm. 10.10	The mechanism is [29, Thm. 10.1]; the statement about jump degrees and the bound (c) are new.
Prop. 10.12	New corollary of [24].
Thm. 10.13	Rewrite of Thm. 4.2 in full generality; ingredients [21, 29].
Lemma 10.14, Cor. 10.15	Extends Prop. 3.9 to non-abelian G ; probably known to experts, we found no reference.
Ex. 10.16	Folklore.
Thm. 10.17	(a),(b),(d) are immediate from the above; (c), i.e. $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$ and $d_{n-2} = 4$, is new (Arason sequence [17] + residue induction).
Ex. 10.19	New observation from [26].
Ex. 10.20	The identity $\text{ed}(\tau_{\text{gen}} \otimes F_m) = \text{ed}(S_{n-m})$ is new but easy; inputs [19, 28].

10.7. Discussion and open questions. In characteristic 0 the profile of τ_{gen} is squeezed between Sylow divisibility (Theorem 10.10: a prime q with $\text{ed}_k(G; q) > j$ divides every $d_i, i \leq j$, and the prime-to- q tail is constant), the roots-of-unity layer (Corollary 10.5: within degree $[k(\mu_e) : k]$, prime to $|G|$ when $|G|$ is a prime power, the profile drops to the representation-theoretic value $\text{ed}_{k(\mu_e)}(G)$), and the subgroup upper bounds (Corollary 10.15). What is unknown is whether non-Galois or non-subgroup extensions can beat the subgroup witnesses. In characteristic p for a p -group (Sections 3–9) there is no Sylow divisibility beyond $p \mid d_0$ and no roots-of-unity layer; the whole profile lives in p -power degrees and records Artin–Schreier–Witt layers. Theorem 10.13 places both regimes in one framework: the profile is the minimal degree of a G -correspondence into a free G -variety of bounded dimension, Theorem 10.10 isolates the part of that degree forced by [29, Thm. 10.1], and the rest, the q -power part in characteristic $\neq q$ and the p -power part in characteristic p , is where the profile carries information beyond ed and $\text{ed}(\cdot; q)$.

Question 10.21. (1) For $(\mathbb{Z}/2)^n$ over an algebraically closed field of characteristic $\neq 2$, is $d_j(\tau_{\text{gen}}) = 2^{n-j}$ for all j (known for $j \in \{0, 1, n-1, n\}$)? For $(\mathbb{Z}/q)^n$ with $\mu_q \subset k$, the cyclic-kernel analogue of the Arason sequence should give $d_{n-1} = q$ [expected]. (2) For $\mathbb{Z}/p$ over $\mathbb{Q}$, is $d_1(\tau_{\text{gen}}) = (p-1)/2$? More generally, when is Corollary 10.5 sharp, and can an extension not of the form Fk' do better than adjoining roots of unity? (3) Determine $d_2(\tau_{\text{gen}}(S_6)) \in \{2, 4, 6\}$; does $d_j(\tau_{\text{gen}}(S_n))$ grow linearly or super-polynomially in n ? (4) Is the profile of $\tau_{\text{gen}}(V)$ independent of the faithful representation V ? (Presumably yes, by specialisation as in [29].)

11. SUPPLEMENTS AT $(p, n) = (2, 2)$ AND $(2, 3)$: THE CUBIC SLOT, QUADRATIC WITNESSES, AND A RIEMANN–ROCH BOUND

Addendum (27 September 2026). This section settles the open item of Theorem 6.2, gives a general Riemann–Roch bound for the prime-to- p jump degree, shows that the Artin–Schreier–Witt curve never carries a quadratic witness at $(2, 3)$ and that a quadratic witness can never have a descent field inside K , produces new families of quadratic extensions with essential dimension exactly 2, and reformulates the cubic slot. Notation is that of Sections 2, 5–7: k algebraically closed of characteristic 2, $K = k(s_0, s_1, s_2)$, $L = k(z_0, z_1, z_2)$, $s = \wp(z)$, $K_2 = k(s_0, s_1)$, τ_2 the generic $\mathbb{Z}/4$ -torsor, $E' : w^2 + w = u(u+1)(u+s_0) + s_1$, $\bar{U}_n$ the Artin–Schreier–Witt curve $\wp y = [t]$, Φ_2 the polynomial of Proposition 7.5 (written out in (4)). We write $(s_0, s_1)_K$ for the $\mathbb{Z}/4$ -class $(s_0, s_1) \in W_2(K)/\wp$, i.e. $\tau_2 \otimes_{K_2} K$; by Lemma 4.1(i), $\text{ed}_k((s_0, s_1)_K) = \text{ed}_k(\tau_2) = 2$. Every separable quadratic extension of K is $K(v_f)$, $v_f^2 + v_f = f$, $f \in K/\wp K$ (Proposition 3.6). Scripts referred to are listed in Appendix F.

11.1. **Supplement to Theorem 6.2: the prime-to-2 jump degree at $n = 2$.**

Theorem 11.1. *Let $p = 2$, $n = 2$. Let u be a root of*

$$P(u) := u^3 + (1 + s_0)u^2 + s_0u + s_1 = u(u+1)(u+s_0) + s_1.$$

Then $F' = K_2(u)$ is a separable cubic extension of K_2 and $(s_0, s_1) \equiv [s_0 + u^2 + u] \bmod \wp W_2(F')$. Hence $\text{ed}_k(\tau_2 \otimes F') = 1$ and

$$d_1^{(p')}(\tau_2) = 3.$$

More precisely, the cubic closed points of E' over K_2 are exactly the sections by the lines $w = \alpha u + \beta$, $\alpha, \beta \in K_2$; each is a single free point of degree 3, and over its residue field (s_0, s_1) is Teichmüller. Consequently $D(U_2) = \mathbb{Z}_{\geq 2}$, and the profile of τ_2 is completely known: $\text{ed}^{[1]} = 2$, $\text{ed}^{[2]} = \text{ed}^{[3]} = 1$, $\text{ed}^{[4]} = 0$, $d_1 = 2$, $d_1^{(p')} = 3$, $d_0 = 4$. [proved]

Proof. $P \in k(s_0)[u][s_1]$ is of degree 1 in s_1 with coprime coefficients 1 and $u(u+1)(u+s_0)$, hence irreducible in $k(s_0)[u, s_1]$, hence (Gauss, P monic in u) irreducible in $K_2[u]$; an irreducible cubic in characteristic 2 is separable. By (3)–(4), for $h = (u, 0) \in W_2(F')$: $(s + \wp h)_0 = s_0 + u^2 + u$ and $(s + \wp h)_1 = s_1 + u^3 + u^2 + s_0u^2 + s_0u = P(u) = 0$, so $s + \wp h = [s_0 + u^2 + u]$; equivalently $(u, 0)$ is an F' -point of $E'^\circ = \tau_2 U_2$ and the last sentence of Theorem 6.2 applies. So $\text{ed}_k(\tau_2 \otimes F') \leq 1$, and ≥ 1 since $3 < 4 = d_0$. As $d_1^{(p')}$ is odd and $\geq d_1(\tau_2) = 2$ (Proposition 6.1), $d_1^{(p')}(\tau_2) = 3$.

Cubic points: $E'(K_2) = \{\infty'\}$ (Theorem 6.2), so $\text{Pic}^0(E')(K_2) = 0$ and every effective K_2 -divisor of degree 3 is linearly equivalent to $3\infty'$, i.e. is the zero divisor of an element of $L(3\infty') = \langle 1, u, w \rangle$; the sections $u = c$ are a degree-2 fibre plus ∞' , so the cubic points are the sections by lines $w = \alpha u + \beta$. Such a section is a single closed point of degree 3 because a K_2 -rational component would be a K_2 -point of E'° . It lies in E'° , hence is free. $D(U_2) \supseteq \{2, 3\}$ and Appendix B gives all of $\mathbb{Z}_{\geq 2}$. $\square$

Remark 11.2. (i) This closes the item marked [open] in Theorem 6.2 and the $p = 2$ part of Question 6.4: the “group-theoretic” obstruction $E'(K_2) = 0$ kills degree-1 points and *produces* the cubic ones. (ii) For odd p the same mechanism gives $d_1^{(p')}(\tau_2) \leq p^2 - p + 1$ (Theorem 11.3); Question 6.4 for degrees in $[2, p - 1]$ is untouched.

11.2. **Supplement to Proposition 3.7 and Remark 4.5: a Riemann–Roch bound.**

Let $l_n = l_n(p)$ be the largest lower ramification break of $\bar{U}_n \rightarrow \mathbb{P}_t^1$ at ∞ , i.e. the pole order at

∞ of the reduced top Witt coordinate of U_n ; by Appendix B,

$$(1) \quad l_n(p) = 1 + (p-1) \sum_{j=1}^{n-1} p^{2j-1} : \quad l_2(2) = 3, \quad l_2(p) = p^2 - p + 1, \quad l_3(2) = 11, \quad l_4(2) = 43.$$

Theorem 11.3. *For every (p, n) , $D(U_n)$ contains an integer prime to p and $\leq l_n$; hence*

$$d_1^{(p')}(\tau_{\text{gen}}) \leq l_n(p).$$

This is the best bound obtainable from the pencils $|d \cdot \infty''|$ on $\tau_{\text{gen}} \bar{U}_n$: every function regular on the affine curve U_n with pole order prime to p has pole order $\geq l_n$. [proved; the closed formula (1) is proved for $(2, 2)$, $(2, 3)$, $(p, 2)$ and cited in general]

Proof. $X := \tau_{\text{gen}} \bar{U}_n$ is a smooth projective geometrically integral K -curve with the K -point ∞'' (twist of the G -fixed point), and its free locus is $X \setminus \{\infty''\}$ (Proposition 3.7(c)). Riemann–Roch dimensions $\ell(d\infty'')$ are invariant under base change to $\bar{K}$, where $(X, \infty'') \cong (\bar{U}_n, \infty)$; so the Weierstrass semigroup of ∞'' on X is that of ∞ on $\bar{U}_n$. The top layer $U_n \rightarrow U_{n-1}$ is $y^p - y = c$ with $v_\infty(c) = -l_n$ on U_{n-1} and ∞ totally ramified, so $v(y) = -l_n$ on U_n and l_n is a pole order. Hence there is $f \in K(X)$ with pole divisor exactly $l_n \infty''$. For $c \in K$ the divisor $\text{div}(f - c) + l_n \infty''$ is effective, K -rational, of degree l_n and supported on $X \setminus \{\infty''\}$. Its closed points have degrees summing to $l_n \equiv 1 \pmod{p}$, so one of them has degree e prime to p , $e \leq l_n$, and it is free; Theorem 4.2 gives $\text{ed}_k^{[e]}(\tau_{\text{gen}}) \leq 1$. For optimality: the coordinate ring of U_n is free over that of U_{n-1} with basis $1, y, \dots, y^{p-1}$, and $v(a_j y^j) = p v_{U_{n-1}}(a_j) - j l_n$ are pairwise distinct modulo p , so a function with pole order prime to p has a dominant term with $j \geq 1$ and pole order $\geq l_n$. $\square$

Corollary 11.4. *At $(p, n) = (2, 3)$: the fibres of the pole-order-11 pencil are free divisors of degree 11 without degree-1 components (Proposition 7.5), so each contains a free point of odd degree in $\{3, 5, 7, 9, 11\}$, and $11 \in D(U_3)$ exactly (Appendix B). Hence*

$$d_1^{(p')}(\tau_{\text{gen}}) \in \{3, 5, 7, 9, 11\}, \quad \delta_{\text{free}}(U_3) \in \{3, 5, 7, 9, 11\},$$

the second by Corollary 11.7 below. This refines “ $d_1^{(p')} \geq 3$ finite” in Theorem 7.1. [proved]

Remark 11.5. The naive argument “ $\ell(d\infty'') \geq 2$ for $d \geq g + 1$, take d odd” does not work: the fibres of $f \in L(d\infty'')$ have degree equal to the actual pole order of f , which lies in the Weierstrass semigroup and may be even. At $(2, 3)$, $g(\bar{U}_3) = 7$ and the semigroup of ∞ is $\langle 4, 6, 11 \rangle$, whose gaps are $\{1, 2, 3, 5, 7, 9, 13\}$: every function of pole order ≤ 10 has even pole order.

11.3. Supplement to Section 7: quadratic witnesses. Fix $f \in K \setminus \wp K$ and $K' = K(v_f)$. A *witness* is such a K' with $\text{ed}_k(\tau_{\text{gen}} \otimes K') = 1$; a *descent field* is a subfield $k \subset F_0 \subset K'$ of transcendence degree 1 with a class $\beta \in W_3(F_0)/\wp$, $s \equiv \beta$ in $W_3(K')/\wp$.

Theorem 11.6 (no descent field inside K). *Let $K' = K(v_f)$ be a witness with descent field F_0 and class β .*

- (a) *If $F_0 \subset K$ then $F_0 = k(t)$ is rational, $s \equiv \beta + \varepsilon(0, 0, f) \pmod{\wp W_3(K)}$ with $\varepsilon \in \{0, 1\}$, and $\text{ed}_k(\tau_{\text{gen}}) \leq \text{trdeg}_k k(t, f) \leq 2$.*
- (b) *The case (a) does not occur: $F_0 \not\subset K$.*
- (c) *The truncation of β is a descent of $(s_0, s_1)_K \otimes K'$, so $f \in \mathfrak{F} := \{f \in K/\wp K : \text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 1\}$.*

[proved]

Proof. (a) $\ker(H^1(K, \mathbb{Z}/8) \rightarrow H^1(K', \mathbb{Z}/8)) = \text{Hom}(\mathbb{Z}/2, \mathbb{Z}/8) \cong \mathbb{Z}/2$ by inflation–restriction, generated by the induced class $V^2[f] = (0, 0, f)$ (Section 2.2). Since s and β both lie in $W_3(K)/\wp$ and agree over K' , $s \equiv \beta + \varepsilon(0, 0, f) = (\beta_0, \beta_1, \beta_2 + \varepsilon f)$ by (2); and a subfield of transcendence degree 1 of $k(s_0, s_1, s_2)$ is $k(t)$ (a curve dominated by $\mathbb{A}^3$ is dominated by a line, Lemma 2.5(i)). (b) Truncate the congruence of (a) to W_2 : the term $(0, 0, f)$ dies and $(s_0, s_1) \equiv (\beta_0, \beta_1) \pmod{\wp W_2(K)}$ with $\beta_0, \beta_1 \in F_0$, so $\text{ed}_k((s_0, s_1)_K) \leq 1$, contradicting $\text{ed}_k((s_0, s_1)_K) = 2$. (c) Truncation commutes with $\wp$ and with base change. $\square$

So a quadratic witness, if one exists, has a descent field *not* contained in K and can never be turned into a counterexample to Ledet’s conjecture by this route; the truncation argument is the one used in the proof of Proposition 7.5 (the step “ $\varepsilon = 1 \Rightarrow$ a K -point of E' ”), applied to an arbitrary descent class instead of $[t]$.

Corollary 11.7 ($\bar{U}_3$ admits no quadratic witness). *For every quadratic K'/K , $\tau_{\text{gen}}\bar{U}_3$ has no free K' -point: s is not Teichmüller over any quadratic extension of K , and $D(U_3) \cap \{1, 2\} = \emptyset$. In particular Algorithm 9.1 has nothing to find, and the “points not coming from the s_2 -line” of Remark 7.6 do not exist.* [proved]

Proof. By Proposition 7.5 a free K' -point needs $Q = (u, w_1) \in E'(K')$ with $u \notin K$. Let σ generate $\text{Gal}(K'/K)$. $Q + \sigma Q$ is σ -invariant, hence lies in $E'(K) = \{\infty'\}$ (Lemma 4.1(ii)); so $\sigma Q = -Q = (u, w_1 + 1)$, whence $\sigma(u) = u$ and $u \in K$. $\square$

The same argument at level 2 shows that $E'(K(v_f)) \neq \{\infty'\}$ iff $f \equiv s_1 + u(u+1)(u+s_0) \pmod{\wp K}$ for some $u \in K$: the E_0 -witnesses of Theorem 6.2 are exactly the u -fibres (Proposition D.2).

Proposition 11.8 (fixed-field trick). *Let $K' = K(v_f)$ with $f \notin \{0, s_0\} + \wp K$, so that $L' = L \otimes_K K' = L(v_f)$ is a field. Suppose there are a subfield $M \subset L'$ and $x \in L'$ with $L' = M(x)$ purely transcendental, such that M is fixed pointwise by some $1 \neq h \in G$. Then $\text{ed}_k(\tau_{\text{gen}} \otimes K') \geq 2$. The same holds for $(s_0, s_1)_K \otimes K'$, with L' replaced by $k(z_0, z_1, s_2)(v_f)$ and G by $\mathbb{Z}/4$.* [proved]

Proof. A witness gives, by Corollary 4.4, a smooth projective curve C with faithful G -action and a G -equivariant dominant rational map from a model of L' ; take the model $B \times \mathbb{A}^1$ with $k(B) = M$. Lemma 7.2 for $B \times \mathbb{A}^1 \rightarrow B$ gives: either a fibre $\mathbb{A}^1$ maps nonconstantly to C , so $C \cong \mathbb{P}^1$ (Lemma 2.5(i)), impossible (Lemma 2.5(ii)); or $k(C) \subset M \subset (L')^h$, so h acts trivially on C , contradicting faithfulness. $\square$

Theorem 11.9 (quadratic extensions with ed exactly 2). $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_f)) = 2$ for every $f \notin \wp K$ in each of the classes

- (a) $f \in K_2 + \wp K$ (this contains $K(v_1)$ and $K(z_0)$ of Theorem 7.3);
- (b) $f \in s_2 + K_2 + \wp K$ (this contains $K(v_2)$, not covered by Theorem 7.3);
- (c) $f \in k(g) + \wp K$ for any $g \in K$ with $K = K_2(g)$, e.g. $f \in k(s_2)$, $f = 1/(s_2 + s_0 s_1)$;
- (d) $f \in k(s_0, s_1)[s_2]$ a polynomial in s_2 of degree ≤ 7 .

In cases (a), (c) already $\text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 2$, i.e. $f \notin \mathfrak{F}$. [proved]

Proof. The upper bound is Theorem 7.1. (a) Replace f by $e \in K_2$ with $f \equiv e$. If $e \equiv s_0$, Proposition 3.9. Otherwise $M := k(z_0, z_1, v_e)$ is algebraic over $k(z_0, z_1)$, $L' = M(z_2)$, and $g^4: z \mapsto z + V^2(1) = (z_0, z_1, z_2 + 1)$ fixes M ; apply Proposition 11.8. (b) Write $f = s_2 + e + \wp h$, $e \in K_2$. Over $K' = K(v_f)$, $s_2 = \wp(v_f + h) + e$, so $s = (s_0, s_1, e) + \wp V^2(v_f + h) \equiv (s_0, s_1, e)$ by (2). Now $K' = K_2(v_f)$ is purely transcendental over K_2 , so by Lemma 4.1(i) $\text{ed}_k(\tau_{\text{gen}} \otimes K') = \text{ed}_k((s_0, s_1, e)_{K_2}) \geq \text{ed}_k((s_0, s_1)_{K_2}) = 2$. (c) Replace f by an element of $k(g)$. Put $F := k(g, v_f)$; then $K' = F(s_0, s_1)$ with s_0, s_1 algebraically independent over F , and the total space of $(s_0, s_1)_K \otimes K'$ is $k(z_0, z_1, g, v_f) = F(z_0, z_1)$, a field on which $\mathbb{Z}/4$ acts fixing F ; the

argument of Proposition 11.8 with the fibration $B \times \mathbb{A}^2 \rightarrow B$ excludes a faithful $\mathbb{Z}/4$ -curve, so $\text{ed}_k((s_0, s_1)_K \otimes K') = 2$, and a descent of s over K' would truncate to a descent of $(s_0, s_1)_K \otimes K'$. (d) If $f \in K_2 + \wp K$ use (a); otherwise Theorem 11.10(b) gives a witness curve of p -rank 0 and genus ≤ 6 , contradicting Corollary 11.11. $\square$

Theorem 11.10 (structure of a quadratic witness). *Let $K' = K(v_f)$ be a witness with $f \notin K_2 + \wp K$, C its curve, $C' = C/G$. For $a \in k^2$ let $D_{f,a}: y^2 + y = f(a_0, a_1, s_2)$ (a curve over k in the s_2 -line) and, for $b \in \wp_2^{-1}(a) \subset \mathbb{A}_z^2$, let D_b be its pullback along $z_2 \mapsto s_2 = z_2^2 + z_2 + C(b_0, b_1)$.*

- (a) *For a in a dense open subset of $\mathbb{A}^2(k)$: $D_{f,a}$ is integral and dominates C' , and every irreducible component of D_b dominates C . Hence $g(C) \leq g(\hat{D}_b)$ and $\sigma(C) \leq \sigma(\hat{D}_b)$ (σ the p -rank, $\hat{D}_b$ the normalisation of a component).*
- (b) *If $f \in k(s_0, s_1)[s_2]$ has s_2 -degree $m \geq 1$, then $\sigma(C) = 0$, $C \rightarrow C'$ is totally ramified at exactly one point and étale elsewhere, C' has p -rank 0 and genus $\leq \lfloor (m-1)/2 \rfloor$, and $7 \leq g(C) \leq m-1$. In particular $m \geq 8$, and for $8 \leq m \leq 15$ one has $C' \cong \mathbb{P}^1$: C is the Artin–Schreier–Witt curve of a polynomial Witt vector $\beta \in W_3(k[t])$, and $s \equiv \beta(t)$ over K' with $t \in K' \setminus K$.*

[proved]

Proof. (a) Let $B_f: y^2 + y = f(s_0, s_1, s_2)$ over $\mathbb{A}_s^3$, $k(B_f) = K'$. Its generic fibre over $\mathbb{A}_{(s_0, s_1)}^2$ is the Artin–Schreier curve $y^2 + y = f$ over K_2 in the s_2 -line, geometrically integral because $f \notin K_2 + \wp K$ (a constant quadratic subextension $K_2(v_e) \subset K'$ would force $f \equiv e$). Apply Lemma 7.2 to $B_f \rightarrow \mathbb{A}_s^3$ and the dominant rational map $B_f \dashrightarrow C'$ given by $k(C') \subset K'$: the alternative $k(C') \subset k(s_0, s_1) \subset K$ is excluded by Theorem 11.6(b), so $D_{f,a} \dashrightarrow C'$ is dominant for general a . Let Y_f be the normalisation of $\mathbb{A}_z^3 \times_{\mathbb{A}_s^3} B_f$, with function field L' ; the equivariant map $Y_f \dashrightarrow C$ composed with $C \rightarrow C'$ factors through $Y_f \rightarrow B_f \dashrightarrow C'$, and the fibre of Y_f over a is $\bigsqcup_{b \in \wp_2^{-1}(a)} D_b$, permuted transitively by G . Each $D_b \rightarrow D_{f,a} \rightarrow C'$ is nonconstant, hence so is $D_b \rightarrow C$. Genus and p -rank do not increase under dominant maps of curves. (b) $D_{f,a}$ is $y^2 + y = (\text{polynomial of degree } m \text{ in } s_2)$, with a single branch point, so $\sigma(D_{f,a}) = 0$ (Deuring–Shafarevich) and $g \leq \lfloor (m-1)/2 \rfloor$; D_b is $y^2 + y = (\text{polynomial of degree } 2m \text{ in } z_2)$, again with one branch point, so $\sigma = 0$ and, after Artin–Schreier reduction to odd degree $\leq 2m-1$, $g(D_b) \leq m-1$. By (a), $\sigma(C) = \sigma(C') = 0$ and $g(C) \leq m-1$; Lemma C.1 and Corollary 11.11 give the ramification structure and $g(C) \geq 7$. If $g(C') = 1$ then Riemann–Hurwitz gives $2g(C) - 2 \geq 0 + 28$, so $m \geq 16$; for $m \leq 15$, $C' = \mathbb{P}^1$ and the cover is étale outside one point, so its class lies in $H^1(\mathbb{A}^1, \mathbb{Z}/8) = W_3(k[t])/\wp$, with β_0 nonconstant since $s_0 \notin \wp K'$ (else $K' = K(z_0)$). $\square$

Corollary 11.11 (minimal genus). *A smooth projective curve with faithful $\mathbb{Z}/8$ -action and p -rank 0 in characteristic 2 has genus ≥ 7 , with equality iff $C/G \cong \mathbb{P}^1$ and the upper ramification breaks at the unique ramified point are $(1, 2, 4)$; $\bar{U}_3$ is an example, $g(\bar{U}_3) = 7$. A faithful $\mathbb{Z}/4$ -curve of p -rank 0 has genus ≥ 1 , with equality iff it is $(E_0, \text{Witt action})$ of Theorem 6.2. [proved]*

Proof. Lemma C.1: one totally ramified point, C/G of p -rank 0. The different exponent at that point is $(u_1 + 1) + 2(u_2 + 1) + 4(u_3 + 1)$ in terms of the upper breaks, with $u_1 \geq 1$ and $u_{i+1} \geq 2u_i$ [23, 31], so $2g - 2 \geq -16 + 28 = 12$. For $\mathbb{Z}/4$: $2g - 2 \geq -8 + 2 + 6 = 0$, and the genus-1 case is (E_0, Witt) by Lemma C.2. $\square$

Remark 11.12. (i) Theorem 7.3's proofs for $K(v_1)$ and $K(v_1, v_2)$ are instances of the mechanism of Theorem 11.10; for $f \in K_2$ the fixed-field trick is shorter, but for the quartic tower it does not apply directly (g^4 moves $y' = v_2 + z_2$) and the argument of Theorem 7.3 is needed. (ii)

The only genus-1 faithful $\mathbb{Z}/8$ -curves are ordinary elliptic curves with G acting by translation by a point of order 8 (Appendix C); by Theorem 11.10(a) they can occur as witness curves only if f has, for general a , at least two poles on the s_2 -line, and then $\tau_{\text{gen}} \otimes K'$ is the étale $\mathbb{Z}/8$ -isogeny torsor of an ordinary elliptic curve at a non-constant point (Appendix D). This and polynomial f of s_2 -degree ≥ 8 are the only remaining shapes of a quadratic witness.

11.4. Supplement to Proposition 7.5 and Question (Q2): the cubic slot. Let F'/K be cubic (separable without loss, Proposition 3.6). Then $L \otimes_K F'$ is a field and Corollary 4.4 applies: a witness for $\text{ed}^{[3]}(\tau_{\text{gen}}) = 1$ is a faithful $\mathbb{Z}/8$ -curve C with a free F' -point on $\tau_{\text{gen}}C$.

Proposition 11.13. *A cubic witness curve C has genus ≥ 3 and, if its p -rank is 0, genus ≥ 7 . Ordinary elliptic curves with translation action carry no free point of odd degree.* [proved]

Proof. $\mathbb{P}^1$ and supersingular elliptic curves are excluded by Lemma 2.5. If C is an ordinary elliptic curve with G acting by translation by P of order 8 (the only genus-1 case, Appendix C), then $\tau_{\text{gen}}C$ is a principal homogeneous space under C whose period divides 8; period and index have the same prime divisors [25], so a nontrivial such torsor has no closed point of odd degree, and a trivial one has a K -point, free since the action is free, forcing $\text{ed}_k(\tau_{\text{gen}}) \leq 1$, false. Genus 2: the hyperelliptic involution is central, and the image of $\mathbb{Z}/8$ in $\text{Aut}(\mathbb{P}^1)$ is $\mathbb{Z}/8$ or $\mathbb{Z}/4$, excluded by Lemma 2.5(ii). p -rank 0: Corollary 11.11. $\square$

On $\bar{U}_3$ the question is completely explicit. By Proposition 5.2(a) and (3)–(4), a free F' -point of $\tau_{\text{gen}}\bar{U}_3$ is $h = (u, w_1, w_2) \in W_3(F')$ with

$$(2) \quad w_1^2 + w_1 = u(u+1)(u+s_0) + s_1, \quad w_2^2 + w_2 = e_Q := s_2 + \Phi_2(s_0, s_1, u, w_1),$$

and then $s \equiv [s_0 + u^2 + u]$ over F' .

Proposition 11.14 (reformulation). $\text{ed}^{[3]}(\tau_{\text{gen}}) = 1$ is witnessed on $\bar{U}_3$ iff there are $\alpha, \beta \in K$ such that, for u a root of

$$u^3 + (1 + s_0 + \alpha^2)u^2 + (s_0 + \alpha)u + (s_1 + \beta^2 + \beta) = 0,$$

one has $e_Q = s_2 + \Phi_2(s_0, s_1, u, \alpha u + \beta) \in \wp K(u)$. Every such cubic is irreducible over K , and these are all the cubic points of E' over K . [proved]

Proof. As in Theorem 11.1, $E'(K) = \{\infty'\}$ (Lemma 4.1(ii)) gives $\text{Pic}^0(E')(K) = 0$, so the cubic points of E' over K are the line sections, each irreducible. A free F' -point with $[F' : K] = 3$ has $F' = K(u)$: if $u \in K$ then $t = s_0 + u^2 + u \in K$ and $s \equiv [t]$ over F' would give $s \equiv [t]$ over K , since $\text{res}_{F'/K}$ is injective on $H^1(\cdot, \mathbb{Z}/8)$ ($\text{cores} \circ \text{res} = 3$ is a unit mod 8). $\square$

Proposition 11.15 (necessary condition and a partial obstruction). (a) *If $s \equiv [t]$ over a cubic $F' = K(u)$ then $s \equiv 3 \cdot \text{cores}_{F'/K}[t]$ in $W_3(K)/\wp$, where cores is the Witt-vector trace $[t] + [t'] + [t''] = (e_1, e_2, e_1e_3 + e_1^2e_2)$ (e_i the elementary symmetric functions of the conjugates). Coordinates 0 and 1 of this condition hold automatically for every line, and coordinate 2 reads*

$$(3) \quad \text{Tr}_{K(u)/K}(e_Q) = s_2 + R(s_0, s_1, \alpha, \beta) \in \wp K$$

with R the explicit polynomial of Appendix E.

- (b) (3) fails for all $\alpha, \beta \in K_2$, and more generally for all α, β with pole order ≤ 1 at $s_2 = \infty$. Hence a cubic witness on $\bar{U}_3$, if any, has a line with α or β of pole order ≥ 2 at $s_2 = \infty$.

[(a) proved, coordinates checked symbolically; (b) proved, with computer-verified coefficients]

Proof. (a) $\text{cores} \circ \text{res} = 3$ on $H^1(K, \mathbb{Z}/8)$ and $3^2 \equiv 1 \pmod{8}$. Corestriction is a morphism of δ -functors for $0 \rightarrow \mathbb{Z}/8 \rightarrow W_3 \xrightarrow{\wp} W_3 \rightarrow 0$, and on H^0 it is the norm, i.e. the Witt trace; the formula for $[t] + [t'] + [t'']$ follows from (3), and the coordinate computations are in Appendix E. (b) For $\alpha, \beta \in K_2$, $s_2 + R$ has a simple pole at $s_2 = \infty$ (Remark 7.4). In general write $\alpha = \sum_{i \leq 1} a_i s_2^i$, $\beta = \sum_{i \leq 1} b_i s_2^i$ in $K_2((1/s_2))$. Membership in $\wp$ of the completion forces, for every odd $i_0 \geq 1$, the chain condition $\sum_j c_{i_0 2^j}^{2^{J-j}} = 0$ on the coefficients c_i of s_2^i in $s_2 + R$ (Appendix E). The only monomial of R producing s_2^{11} from exponents ≤ 1 is α^{11} , giving $c_{11} = a_1^{11}$ and $a_1 = 0$; then $c_3 = b_1^3$ gives $b_1 = 0$; then α, β are integral at ∞ and $s_2 + R$ has a simple pole, contradicting the chain condition for $i_0 = 1$. $\square$

Exhaustive searches over $\mathbb{F}_2$ -polynomial lines (about $2.4 \cdot 10^6$ pairs (α, β) of small degree, Appendix E) found no line satisfying (3). Note that (3) is only necessary: the kernel of cores on $H^1(K(u), \mathbb{Z}/2)$ is large. Our expectation is that no cubic witness exists on $\bar{U}_3$; whether another faithful $\mathbb{Z}/8$ -curve of genus ≥ 3 carries one is a separate question, so (Q2) remains open.

11.5. Updated status at (2, 3) and revised questions.

Theorem 11.16 (status at (2, 3)).

$$\text{ed}^{[1]} \in \{2, 3\}, \quad \text{ed}^{[2]} \in \{1, 2\}, \quad \text{ed}^{[3]} \in \{1, 2\}, \quad \text{ed}^{[4]} = \dots = \text{ed}^{[7]} = 1, \quad \text{ed}^{[8]} = 0;$$

$d_2 \in \{1, 2\}$, $d_1 \in \{2, 3, 4\}$, $d_0 = 8$, $d_1^{(p')} \in \{3, 5, 7, 9, 11\}$. Moreover: a quadratic witness for $d_1 = 2$ has no descent field inside K , is not $\bar{U}_3$, is not any f of Theorem 11.9, and for polynomial f is an Artin–Schreier–Witt curve of a polynomial Witt vector of s_2 -degree ≥ 8 ; $D(U_3) \cap \{1, 2\} = \emptyset$, $D(U_3) \supseteq \langle 4, 6, 11 \rangle \setminus \{0\}$, and $D(U_3) \cap \{3, 5, 7, 9, 13\}$ is unknown. [proved]

The open questions of Section 9.2 are refined as follows. (Q1): is there $f \in K/\wp K$, with at least two poles on the s_2 -line for general (s_0, s_1) or a polynomial of s_2 -degree ≥ 8 , such that $K(v_f)$ is a witness? The first shape requires the étale $\mathbb{Z}/8$ -isogeny torsor of an ordinary elliptic curve (Appendix D); the level-2 version (is $(s_0, s_1)_K \otimes K(v_f)$ the étale $\mathbb{Z}/4$ -isogeny torsor of an ordinary elliptic curve at a non-constant point?) is the natural first step, since a negative answer there kills all such level-3 witnesses by truncation. (Q2): does (3) fail for all $\alpha, \beta \in K$ (a local analysis at $s_2 = \infty$ for pole orders ≥ 2 is the missing piece), and which of 3, 5, 7, 9 lie in $D(U_3)$? The splitting type over K of the degree-11 fibres of the pole-order-11 pencil is a finite computation that would decide the latter. (Q3), (Q4) are unchanged, except that $d_1^{(p')}(\tau_2) \leq p^2 - p + 1$ for all p and $= 3$ for $p = 2$.

APPENDIX A. WITT-VECTOR COMPUTATIONS AT $p = 2$, $n = 3$

All formulas were re-derived from the ghost components $w_0 = x_0$, $w_1 = x_0^2 + 2x_1$, $w_2 = x_0^4 + 2x_1^2 + 4x_2$ over $\mathbb{Z}$ and reduced modulo 2 (script `witt_check.py`); they agree with (3)–(4). [computed]

Negation. $-(a_0, a_1, a_2) = (a_0, a_1 + a_0^2, a_2 + a_1^2 + a_0^2 a_1 + a_0^4)$; in particular $-x \neq x$, but $\{s - \wp x\} = \{s + \wp x\}$ as x ranges over $W_3(K)$.

The general translate. For $x = (x_0, x_1, x_2) \in W_3(K)$, $s + \wp x = (t, r_1, r_2)$ with

$$\begin{aligned} t &= s_0 + x_0^2 + x_0, \\ r_1 &= s_1 + x_0(x_0 + 1)(x_0 + s_0) + \wp(x_1), \\ r_2 &= s_2 + \Phi_2(s_0, s_1, x_0, x_1) + \wp(x_2), \end{aligned}$$

where

$$(4) \quad \begin{aligned} \Phi_2 = & s_0^3(x_0^2 + x_0) + s_0s_1(x_0^2 + x_0) + s_0(x_0^6 + x_0^4) + s_0(x_0^2 + x_0)(x_1^2 + x_1) \\ & + s_1(x_0^3 + x_0^2) + s_1(x_1^2 + x_1) + x_0^7 + x_0^4 + (x_0^3 + x_0^2)(x_1^2 + x_1) + x_1^3 + x_1^2. \end{aligned}$$

Thus the u -fibre family of Theorem 6.2 is the x_1 -independent part of r_1 : $(s + \wp(c, x_1))_1 - (c(c + 1)(c + s_0) + s_1) = \wp(x_1)$, and $(s - \wp x)_1 - (s + \wp x)_1 = x_0^4 + x_0^2 \in \wp K$.

APPENDIX B. GENUS, RAMIFICATION AND WEIERSTRASS SEMIGROUP OF $\bar{U}_n$

Ramification at ∞ . $\bar{U}_n \rightarrow \mathbb{P}_t^1$ is a $\mathbb{Z}/p^n$ -cover, étale over $\mathbb{A}^1$ and totally ramified at ∞ , of Witt class $[t]$ with $v_\infty(t) = -1$. By the conductor formula for Artin–Schreier–Witt extensions [31] (upper breaks $u_j = \max_{i < j} p^{j-1-i}m_i$ for reduced pole orders m_i) the upper breaks are $u_j = p^{j-1}$, and Herbrand’s function converts them to the lower breaks

$$l_1 = 1, \quad l_{j+1} = l_j + p^j(u_{j+1} - u_j) = l_j + (p-1)p^{2j-1},$$

which gives (1).

[cited] Direct verification at (2, 3) [computed, `genus_U3.py`]: the tower $U_3 \rightarrow U_2 = E_0 \rightarrow \mathbb{P}_{y_0}^1 \rightarrow \mathbb{P}_t^1$ has layers $y_0^2 + y_0 = t$ (break 1), $y_1^2 + y_1 = y_0^3 + y_0^2$ (break 3) and $y_2^2 + y_2 = C(y_0, y_1)$ with C from (4). On E_0 , using $y_1^2 = y_1 + y_0^3 + y_0^2$, $C = y_0^7 + y_0^6 + (y_0^3 + y_0^2)y_1$, of pole order 14; one reduction step, $C + \wp(y_0^2y_1) = y_0^4y_1 + y_0^3y_1$, has odd pole order $8 + 3 = 11$. So the lower breaks are (1, 3, 11) and the upper breaks (1, 2, 4), as predicted.

Genus. With $l_0 := -1$, Riemann–Hurwitz gives $2g(\bar{U}_n) - 2 = -2p^n + \sum_{j=1}^n (p^{n-j+1} - 1)(l_j - l_{j-1})$, whence

$$g(\bar{U}_2) = \frac{1}{2}p(p-1)^2, \quad g(\bar{U}_3) = 7 \ (p=2), \quad g(\bar{U}_4) = 35 \ (p=2), \quad g(\bar{U}_3) = 78 \ (p=3).$$

Cross-check at (2, 3): the different exponent is $(u_1 + 1) + 2(u_2 + 1) + 4(u_3 + 1) = 2 + 6 + 20 = 28$ and $2g - 2 = -16 + 28 = 12$. [computed]

Weierstrass semigroup at (2, 3). The functions $y_0, y_1, y_2 + y_0^2y_1$ are regular on U_3 with pole orders 4, 6, 11 at ∞ ; $\langle 4, 6, 11 \rangle$ has exactly the 7 = g gaps $\{1, 2, 3, 5, 7, 9, 13\}$, so it is the full semigroup of ∞ . The smallest odd pole order of a function regular on U_3 is therefore 11.

Degrees realised on $\bar{U}_n$. Let H be the semigroup of ∞ on $\bar{U}_n$. For $d \in H$ choose $f \in K(\tau_{\text{gen}}\bar{U}_n)$ with pole divisor $d\infty''$ and $df \neq 0$ (possible for every $d \in H$ prime to p , and at (2, 3) for every $d \in \langle 4, 6, 11 \rangle$, e.g. $y_0, y_1, y_0^2 + y_1, y_0y_1$ for 4, 6, 8, 10). Then $K(\tau_{\text{gen}}\bar{U}_n)$ is a separable extension of degree d of $K(f)$, and since $K = k(s_0, \dots, s_{n-1})$ is Hilbertian [22], for infinitely many $c \in K$ the fibre $f = c$ is a single closed point of degree d in the free locus. Hence

$$D(U_n) \supseteq \{d \in H \setminus \{0\} : d \text{ prime to } p\}, \quad D(U_3) \supseteq \langle 4, 6, 11 \rangle \setminus \{0\}, \quad D(U_2) = \mathbb{Z}_{\geq 2} \ (p=2).$$

With Corollary 11.7, $D(U_3) \cap \{1, 2\} = \emptyset$; the unknown part of $D(U_3)$ is $\{3, 5, 7, 9, 13\}$, and the odd degrees realised on $\bar{U}_3$ so far are the odd elements ≥ 11 of $\langle 4, 6, 11 \rangle$ (13 is a gap). [proved, modulo Hilbert irreducibility]

APPENDIX C. CYCLIC 2-POWER ACTIONS ON CURVES IN CHARACTERISTIC 2

Let C be a smooth projective curve over k with a faithful action of $G = \mathbb{Z}/2^n$, $\sigma(\cdot)$ the p -rank. For a G -orbit with inertia of order 2^a and upper breaks $u_1 < \dots < u_a$ one has $u_1 \geq 1$ and $u_{i+1} \geq 2u_i$ [23, 31]; the different exponent at a point of that orbit is $\sum_{i=1}^a (2^i - 2^{i-1})(u_i + 1)$ in terms of the upper breaks of the inertia group. Deuring–Shafarevich [20]:

$$\sigma(C) - 1 = 2^n(\sigma(C/G) - 1) + \sum_{\text{orbits}} 2^n \left(1 - \frac{1}{e_{\text{orbit}}}\right).$$

Lemma C.1. *If $\sigma(C) = 0$ then $\sigma(C/G) = 0$, and $C \rightarrow C/G$ is totally ramified at exactly one point and étale elsewhere. Conversely such a cover of a p -rank-0 curve has p -rank 0. [proved]*

Proof. $2^n \sigma(C/G) + \sum 2^n(1 - 1/e) = 2^n - 1$; each orbit with $e \geq 2$ contributes at least 2^{n-1} , and $2^n - 1$ is reached only by a single orbit with $e = 2^n$. $\square$

Lemma C.2. *Let E be the supersingular elliptic curve over k . Every faithful $\mathbb{Z}/4$ -action on E is conjugate in $\text{Aut}(E)$ to one fixing O , and the six elements of order 4 of $\text{Aut}(E, O) \cong \text{SL}_2(\mathbb{F}_3)$ form one conjugacy class; so up to isomorphism of G -curves (and replacing the generator by its inverse) $(E, \mathbb{Z}/4)$ is $(E_0, \text{Witt action})$. There is no faithful $\mathbb{Z}/8$ -action on E (Lemma 2.5(iii)). [proved]*

Proof. Write $\varphi = t_P \circ \alpha$ with $\alpha \in \text{Aut}(E, O)$. If $\alpha = 1$, $\text{ord } \varphi = \text{ord } P$ is odd ($E(k)[2] = 0$); if $\alpha = -1$, $\varphi^2 = t_{P+\alpha P} = 1$. So order 4 forces $\text{ord } \alpha = 4$, and $t_Q \varphi t_{-Q} = t_{(1-\alpha)Q} \alpha$ with $1 - \alpha$ a nonzero isogeny, surjective on $E(k)$; choose Q with $(1 - \alpha)Q = -P$. The conjugacy classes of $\text{SL}_2(\mathbb{F}_3)$ are $\{1\}$, $\{-1\}$, two of order 3, two of order 6 and one of order 4, of size 6. $\square$

Enumeration (`rh_enum.py`, output `rh_enum.out`) [computed]. Imposing only the necessary conditions above (integrality and non-negativity of genus and p -rank at every level of the tower $C \rightarrow C/\langle g^{2^{n-1}} \rangle \rightarrow \cdots \rightarrow C/G$, Hasse–Arf, $u_{i+1} \geq 2u_i$), the possible $(g(C), g(C/G), \sigma(C/G), \sigma(C))$ for faithful $\mathbb{Z}/8$ -curves with $g(C) \leq 10$ are:

$g(C)$	C/G	ramification	$\sigma(C)$
1	ordinary elliptic	free action (translation by an 8-torsion point)	1
5	ordinary elliptic	one orbit with inertia $\mathbb{Z}/2$	5
7	$\mathbb{P}^1$	one totally ramified point, lower breaks (1, 3, 11)	0
9	$\mathbb{P}^1$	one totally ramified point, lower breaks (1, 3, 15)	0
9	genus 1 or 2	various (one or two orbits, or free)	5, 7, 9 or 1

Consequences: a faithful $\mathbb{Z}/8$ -curve of genus ≤ 6 is an ordinary elliptic curve with translation action or has genus 5 and p -rank 5; no faithful $\mathbb{Z}/8$ -curve has genus 2, 3, 4, 6, 8; and a p -rank-0 one of genus ≤ 10 has genus 7 or 9, with $C/G = \mathbb{P}^1$, $\mathbb{Z}/4$ -quotient a supersingular elliptic curve and $\mathbb{Z}/2$ -quotient $\mathbb{P}^1$. For $\mathbb{Z}/4$ and $g(C) \leq 6$: the p -rank-0 entries have a single totally ramified point with lower breaks $(1, 2j+1)$ (genus j) or C/G of genus 1; the only other genus-1 entry is an ordinary elliptic curve with translation by a 4-torsion point. A genus-2 faithful $\mathbb{Z}/4$ -curve has p -rank 0, quotient $\mathbb{P}^1$ and lower breaks $(1, 5)$.

APPENDIX D. THE LEVEL-2 SET $\mathfrak{F}$ AND THE ISOGENY MECHANISM

Recall $\mathfrak{F} = \{f \in K/\wp K : \text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 1\}$ (Theorem 11.6(c)). Call a descent of $(s_0, s_1)_K \otimes K(v_f)$ “of Case A” if its descent field lies inside K .

Proposition D.1 (level-2 Case A is explicit). *f admits a level-2 Case-A descent, i.e. $(s_0, s_1) \equiv \beta(t)$ over $K(v_f)$ with $t \in K$ and $\beta \in W_2(k(T))$, iff*

$$f \equiv s_1 + x_0(x_0 + 1)(x_0 + s_0) + \beta_1(t) \pmod{\wp K}$$

for some $x_0, t \in K$ and $\beta_0, \beta_1 \in k(T)$ with $\beta_0(t) = s_0 + x_0^2 + x_0$. This set $\mathfrak{F}_A$ contains $\{s_1 + x_0(x_0 + 1)(x_0 + s_0) + b(s_0 + x_0^2 + x_0) : x_0 \in K, b \in k(T)\}$ (the case $\beta_0 = T$). [proved]

Proof. “ $\Leftarrow$ ”: over $K' = K(v_f)$ put $x_1 := v_f + h$ where $f = s_1 + x_0(x_0 + 1)(x_0 + s_0) + \beta_1(t) + \wp h$; by Appendix A, $(s_0, s_1) + \wp(x_0, x_1) = (\beta_0(t), \beta_1(t))$. “ $\Rightarrow$ ”: the descent field is $k(t)$ (Theorem 11.6(a)); (s_0, s_1) and $\beta(t)$ agree over K' , so $(s_0, s_1) \equiv \beta(t) + \varepsilon(0, f)$ over K with $\varepsilon = 1$ since $\text{ed}_k((s_0, s_1)_K) = 2$; read off the coordinates of $(s_0, s_1) + \wp(x_0, x_1) = (\beta_0(t), \beta_1(t) + f)$. $\square$

Proposition D.2 (the E_0 -part of $\mathfrak{F}$). $\{f : E'(K(v_f)) \neq \{\infty'\}\} = \{s_1 + u(u+1)(u+s_0) : u \in K\} + \wp K \subset \mathfrak{F}_A$. Equivalently, the quadratic twist $E'^{(f)} : w^2 + w = u(u+1)(u+s_0) + s_1 + f$ has an affine K -point iff f is in this family, and never one with $u \notin K$. [proved]

Proof. As in Corollary 11.7, every $Q \in E'(K')$ has $\sigma Q = -Q$, so $u(Q) \in K$; then $w \in K' \setminus K$ with $w^2 + w \in K$ forces $w = a + v_f$, $a \in K$, and $f \equiv s_1 + u(u+1)(u+s_0)$. Conversely such f gives the point $(u, v_f + a)$. $\square$

Proposition D.3 ($f \in k(s_0)$). For $e \in k(s_0)$ with $e \notin \{0, s_0\} + \wp k(s_0)$: $\text{ed}_k(\tau_2 \otimes K_2(v_e)) = 2$. Hence $k(s_0) \cap \mathfrak{F} = \{s_0\} + \wp K$. [proved]

Proof. The total space of $\tau_2 \otimes K_2(v_e)$ is $k(z_0, v_e)(z_1) = M(z_1)$ with $M = k(z_0, v_e)$ fixed by $g^2 : (z_0, z_1) \mapsto (z_0, z_1 + 1)$; apply Proposition 11.8 for $\mathbb{Z}/4$. (No genus hypothesis on the curve $v^2 + v = e(z_0^2 + z_0)$ is needed; for $e \in k(s_0)$ the twist $E'^{(e)}$ is isomorphic to E' via $s_1 \mapsto s_1 + e$, which shows directly $E'^{(e)}(K_2) = 0$, consistent with Proposition D.2.) $\square$

Proposition D.4 (corestriction exclusion of constant descents). Let $M_0 \subset K$ be a subfield algebraically closed in K , $f' \in M_0$ with $f \equiv f' \pmod{\wp K}$, $M = M_0(v_{f'})$, so $K' = K \otimes_{M_0} M$. If $s_0 \notin M_0 + \wp K$ (e.g. $M_0 \subset k(s_1, s_2)$), then no descent field of $(s_0, s_1)_K \otimes K'$, in particular none of $\tau_{\text{gen}} \otimes K'$, is contained in M . [proved]

Proof. Let $(s_0, s_1) \equiv \beta$ over K' with $\beta \in W_2(M)/\wp$. Corestriction is compatible with base change along $M_0 \rightarrow K$, so $\text{cores}_{K'/K}(\beta) = \gamma$ comes from $W_2(M_0)/\wp$, while $\text{cores}_{K'/K} \text{res}(s_0, s_1) = 2(s_0, s_1) = VF(s_0, s_1) = (0, s_0^2) \equiv (0, s_0)$. So $(0, s_0) \equiv (\gamma_0, \gamma_1)$ over K ; $\gamma_0 \in \wp K \cap M_0 = \wp M_0$, so after modifying γ by $\wp(h, 0)$, $\gamma_0 = 0$ and $s_0 + \gamma_1 \in \wp K$ by (5), contradicting $s_0 \notin M_0 + \wp K$. $\square$

This does not exclude a general Case-B descent: for a σ -stable descent field it only says that $(0, s_0) = \text{Ind}(s_0)$, which already has $\text{ed} = 1$, descends to it.

The isogeny mechanism. By Theorem 11.10(a) and Appendix C, a witness curve of genus ≤ 6 is an ordinary elliptic curve C with G acting by translation by a point P of order 8, so $C' = C/\langle P \rangle$ is ordinary elliptic, $C \rightarrow C'$ is the étale $\mathbb{Z}/8$ -isogeny (the dual of V^3 , i.e. $C = C'^{(8)}$), and C' is dominated by $D_{f,a}$ for general a : the Jacobians of the curves $D_{f,a}$ must have a fixed ordinary elliptic factor, which requires $f(a, s_2)$ to have at least two poles on the s_2 -line. The class of $\tau_{\text{gen}} \otimes K'$ is then $\delta(Q)$ for the connecting map $\delta : C'(K') \rightarrow H^1(K', \mathbb{Z}/8)$ of $0 \rightarrow \mathbb{Z}/8 \rightarrow C \rightarrow C' \rightarrow 0$ and the non-constant point $Q \in C'(K')$ given by $k(C') \subset K'$. For $n = 1$ and $C' : y^2 + xy = x^3 + a_6$ (ordinary, $j = 1/a_6$), the substitution $y = xz$ gives $z^2 + z = x + a_6/x^2$, so the étale $\mathbb{Z}/2$ -isogeny torsor at (x, y) has Artin-Schreier class $x + a_6/x^2$; the $\mathbb{Z}/8$ version is the Witt vector of the tower $C'^{(8)} \rightarrow C'^{(4)} \rightarrow C'^{(2)} \rightarrow C'$, not computed here. Since δ is a homomorphism and s is σ -invariant, $\delta(Q + \sigma Q) = 2s \equiv (0, 0, s_0)$ and $\delta(Q - \sigma Q) = 0$, i.e. $Q - \sigma Q$ lifts to $C(K')$. We have not derived a contradiction from these constraints. [open]

APPENDIX E. THE CUBIC SLOT ON $\bar{U}_3$: WITT TRACE, CHAIN CONDITIONS, SEARCHES

Witt trace. For a cubic $u^3 + Au^2 + Bu + C$ with $A = 1 + s_0 + \alpha^2$, $B = s_0 + \alpha$, $C = s_1 + \beta^2 + \beta$ and $t = s_0 + u^2 + u$, the trace $[t] + [t'] + [t'']$ has coordinates $(e_1, e_2, e_1e_3 + e_1^2e_2)$ in the elementary symmetric functions of t, t', t'' (from (3); the ghost components of $\sum [t_i]$ are the power sums $\sum t_i^{2^m}$). Moreover $3x = (x_0, x_1 + x_0^2, x_2 + x_1^2 + x_0^2x_1)$. The scripts `cores_cubic*.py` compute $d := 3s - \text{Tr}[t]$: $d_0 = \wp(A)$; after subtracting $\wp[A]$ the coordinate 1 is $\wp(\alpha^4 + \alpha^3 + \alpha s_0 + \beta + s_0^2 + s_0)$, automatically in $\wp K$ (the $n = 2$ statement of Theorem 11.1 seen through `cores`); after subtracting $\wp V[y_1]$ the last coordinate is $s_2 + R(s_0, s_1, \alpha, \beta)$ with R a polynomial of 67 terms,

and it agrees with $\text{Tr}_{K(u)/K}(e_Q)$ modulo $\wp K$. Reducing squares ($x^2 \equiv x$), $R \equiv R_{\text{red}}$ with

$$\begin{aligned} R_{\text{red}} = & \alpha^{11} + s_0\alpha^{10} + s_0\alpha^9 + \alpha^8\beta + s_1\alpha^8 + s_0\alpha^7 + \alpha^6\beta + s_1\alpha^6 + s_0\alpha^6 + \alpha^5\beta^2 + \alpha^5\beta + \alpha^5 + s_1\alpha^5 + s_0^2\alpha^5 \\ & + s_0\alpha^4\beta^2 + \alpha^4\beta + s_0\alpha^4\beta + s_0s_1\alpha^4 + \alpha^3\beta + \alpha^3 + s_0\alpha^3 + s_0^3\alpha^3 + s_0^4\alpha^3 + s_0^2\alpha^2\beta + s_0\alpha^2 + s_0^2s_1\alpha^2 \\ & + s_0^3\alpha^2 + s_0^5\alpha^2 + \alpha\beta^2 + s_0\alpha\beta^2 + s_0^2\alpha\beta^2 + \alpha\beta + s_0^2\alpha\beta + \alpha + s_1\alpha + s_0\alpha + s_0s_1\alpha + s_0^2s_1\alpha + s_0^4\alpha \\ & + s_0^5\alpha + \beta^3 + s_1\beta^2 + s_0\beta^2 + s_0^3\beta^2 + \beta + s_1\beta + s_0^3\beta + s_0^4\beta + s_1 + s_0s_1 + s_0^2s_1 + s_0^3s_1 + s_0^4s_1. \end{aligned}$$

The twisted G -action on W_3 acts on lines by $(\alpha, \beta) \mapsto (\alpha + 1, \alpha + \beta + 1)$, $(\alpha, \beta + 1)$ and preserves (3) (checked symbolically). [computed]

Chain conditions at $s_2 = \infty$. Let $\hat{K} = K_2((1/s_2))$ and $e = \sum_{i \leq N} c_i s_2^i \in \hat{K}$, $c_i \in K_2$. Then $e \in \wp \hat{K}$ iff $c_0 \in \wp K_2$ and, for every odd $i_0 \geq 1$, with $2^J i_0$ the last index of the chain $i_0, 2i_0, 4i_0, \dots$ carrying a nonzero coefficient,

$$(\text{Ch}_{i_0}) \quad \sum_{j=0}^J c_{i_0 2^j}^{2^{J-j}} = 0 \quad \text{in } K_2.$$

(The negative part is always $\wp$ -exact; the solution $y = \sum b_i s_2^i$ of $y^2 + y = \sum_{i>0} c_i s_2^i$ has $b_{i_0} = c_{i_0}$, $b_{2m} = c_{2m} + b_m^2$, and must be a finite sum.) In particular a top coefficient $c_N \neq 0$ with N odd excludes $e \in \wp K$. Proposition 11.15(b) uses: for $\alpha = a_0 + a_1 s_2$, $\beta = b_0 + b_1 s_2$ (plus arbitrary tails of negative order), $c_{11} = a_1^{11}$ and $c_{13} = c_{14} = c_{15} = 0$; with $a_1 = 0$, $c_3 = b_1^3$ and $c_6 = c_{12} = 0$ (`deg1_family.py`). For pole orders ≥ 2 the chain equations are nonlinear in the coefficients and Laurent tails enter; no general argument is known to us.

Searches (`f2poly.py`, `search_lines.py`, `search_targeted.py`) [computed]. For $\alpha, \beta \in \mathbb{F}_2[s_0, s_1, s_2]$, membership $s_2 + R \in \wp K$ is equivalent to membership in $\wp(k[s_0, s_1, s_2])$ and is decided exactly by the monomial-chain algorithm (top monomial must be a square; subtract $\wp$ of its root; iterate). None of the following families contains a line satisfying (3): $\deg \alpha \leq 1$, $\deg \beta \leq 2$ (16384 lines); $\alpha = 0$, $\deg \beta \leq 3$ (2^{20} lines); $\alpha \in \mathbb{F}_2[s_0, s_1]$ of degree ≤ 2 with β of s_2 -degree ≤ 2 and (s_0, s_1) -degree ≤ 1 (2^{15}); α of s_2 -degree ≤ 1 and (s_0, s_1) -degree ≤ 1 with β of s_2 -degree ≤ 2 resp. ≤ 3 (2^{15} resp. 2^{18}); α, β both of total degree ≤ 2 (2^{20}). A finite-field specialisation test (`ff_test.py`: specialise $(s_0, s_1, s_2) \in \mathbb{F}_8^3$, take the roots $u \in \mathbb{F}_{512}$ of the cubic and compute the absolute trace of $e_Q(u)$) is available as an independent check of the full condition $e_Q \in \wp K(u)$ on any candidate; it has not been needed.

APPENDIX F. SCRIPTS

All scripts are Python 3 and live in `notes/scripts/` of the repository: `witt_check.py` (ghost-component verification of (3)–(4), prints Φ_2), `wittlib.py`, `ed2_witt.py` (negation, $s \pm \wp x$), `genus_U3.py` (Artin–Schreier reduction on E_0 , $g(\bar{U}_3) = 7$), `rh_enum.py` (ramification enumeration, output `rh_enum.out`), `cores_cubic.py`, `cores_cubic2.py`, `cores_cubic3.py`, `resolvent.py` (Witt trace, R , invariance under the twisted action), `build_R.py`, `f2poly.py`, `search_lines.py`, `search_targeted.py`, `deg1_family.py`, `ff_test.py`.

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